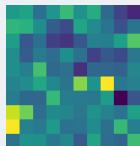


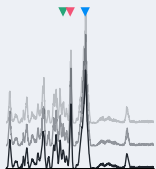
Input



SERS map

DQ24 : TBZ12 : THI6 μM

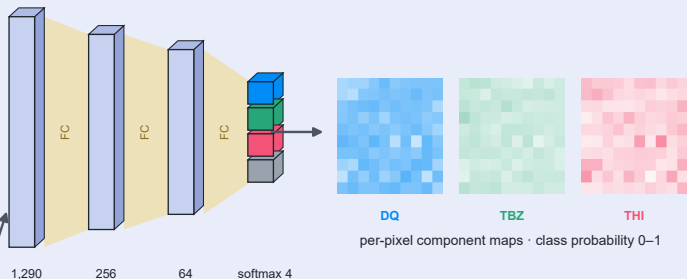
↓ NNLS hit gate



hit-pixel spectra ($n = 100$)

$\log(1+x) \cdot 1,290$ channels
500–2500 cm^{-1}

Composition module — shared per-pixel MLP



↓ probability-weighted band signal I_{eq} (1570 / 1270 / 1367 cm^{-1})

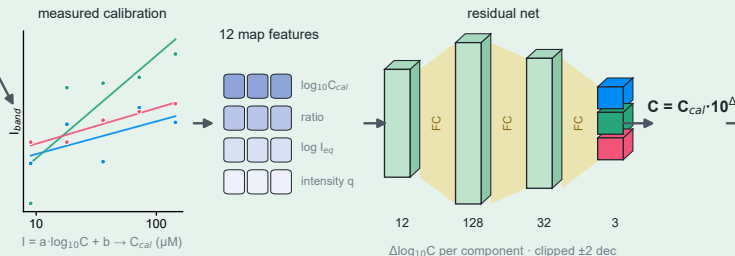
map composition



mean pool over hit pixels
BLK removed · renormalised



Quantification module — calibration-anchored residual



Output

concentration (μM)

C_{DQ} C_{TBZ} C_{THI}

validated window · OOD flags · range-clipped

known total (optional): $C_i = p_i \times C_{total}$

declared metadata — constrained